

PAPER_166 — UQFF Solar Wind Modulation: ϵ_{sw} and Wind-Modified Buoyancy

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Abstract

This paper establishes the solar wind modulation factor in the UQFF Ubi buoyancy terms. A new coupling parameter $\epsilon_{sw} = 0.001$ (buoyancy solar wind factor) scales the solar wind density ρ_{sw} to produce a multiplicative correction $wind_mod = 1 + \epsilon_{sw} \cdot \rho_{sw}$ on each buoyancy term. This extends the Ug2 δ_{sw} term (which enters multiplicatively) with a consistent physical model across all four buoyancy Ubi terms.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Background — Buoyancy Terms in UQFF

The UQFF buoyancy force (Ubi):

$$U_{b,i} = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot (1 + \delta_{sw} v_{sw}) \cdot U_{UA} \cdot \cos(\pi t_n)$$

where $\delta_{sw} \cdot v_{sw}$ is the solar wind velocity modulation in Ug2. This term was previously inconsistent with the wind correction in Ug2 which used $(1 + \delta_{sw} \cdot v_{sw})$ with dimensional mismatch when v_{sw} was given in m/s vs. the normalized wind density.

2. Wind Modulation Factor (NEW)

$$wind_mod = 1 + \epsilon_{sw} \cdot \rho_{sw}$$

Parameter	Value	Units	Physical Basis
ϵ_{sw}	0.001	m^3/kg	Buoyancy solar wind coupling factor
ρ_{sw}	$\sim 5 \times 10^{-21}$	kg/m^3	Solar wind density at 1 AU (proton density $\sim 5/\text{cc}$)

At 1 AU ($\rho_{sw} = m_p \times 5e6 \text{ m}^{-3} = 8.35 \times 10^{-21} \text{ kg}/\text{m}^3$):

$$wind_mod = 1 + 0.001 \times 8.35 \times 10^{-21} = 1.0000000000000000000000084$$

→ The correction is $\sim 10^{-23}$ at 1 AU — negligibly small in the Solar System but significant at stellar wind compressed regions ($r \ll 1 \text{ AU}$, $\rho \gg \rho_{sw}(1 \text{ AU})$).

3. Corrected Ubi Equation

$$U_{b,i}(r, t) = -\beta_i \cdot U_{gi}(r, t) \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot wind_mod \cdot U_{UA} \cdot \cos(\pi t_n)$$

where $wind_mod = 1 + \epsilon_{sw} \cdot \rho_{sw}(r)$ and $\rho_{sw}(r)$ falls off as:

$$\rho_{sw}(r) = \rho_{sw,0} \cdot \left(\frac{r_0}{r}\right)^2$$

at $r_0 = 1$ AU (density decreases as inverse square with distance).

4. Extended Buoyancy — H_SCm Integration

The superconductive medium height H_{SCm} (PAPER_064 canonical $H_{SCm} \approx 0.99$) also enters the buoyancy via:

$$U_{b,i} \propto H_{SCm} \cdot wind_mod$$

For $H_{SCm} = 0.99$ (99% SCm coverage):

$$U_{b,i}(H_{SCm}) = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot M_{bh}/d_g \cdot H_{SCm} \cdot wind_mod \cdot U_{UA} \cdot \cos(\pi t_n)$$

5. Physical Mechanism

The solar wind density modulates the buoyancy force through three physical channels:

1. **Plasma density effect:** Higher ρ_{sw} increases the ambient medium density, which increases the buoyant uplift (Archimedes principle: $F_b \propto \rho_{fluid} \times V \times g$)
2. **Ram pressure effect:** Solar wind ram pressure $P_{ram} = \rho_{sw} v_{sw}^2/2$ compresses the magnetosphere boundary, altering the effective R_b and thus U_g
3. **Ion-neutral friction:** Wind ions interact with neutral UQFF vacuum, transferring momentum $\rightarrow$ drift term in Ubi

6. Solar Wind Density at Different Radii

Location	r/AU	ρ_{sw} [kg/m ³]	wind_mod
Solar corona	0.01	8.35×10^{-17}	$1 + 8.35 \times 10^{-20}$
Mercury	0.39	5.48×10^{-19}	$1 + 5.48 \times 10^{-22}$
Earth (1 AU)	1.0	8.35×10^{-21}	$1 + 8.35 \times 10^{-24}$
Jupiter (5 AU)	5.2	3.09×10^{-22}	$1 + 3.09 \times 10^{-25}$
Termination	100	8.35×10^{-25}	$1 + 8.35 \times 10^{-28}$

The wind_mod correction only becomes dynamically significant ($>1\%$) at $\rho_{sw} > 10^3$ kg/m³, which corresponds to conditions inside accretion disks or dense stellar winds.

7. Consistency with Ug2 (δ_{sw} Parameter)

The Ug2 term used $\delta_{sw} \cdot v_{sw}$ with $\delta_{sw} = 0.001$ and $v_{sw} = 400$ km/s:

$$1 + \delta_{sw} v_{sw} = 1 + 0.001 \times 4 \times 10^5 = 1.4 \quad (40\% \text{ correction})$$

The new wind_mod is consistent when ρ_{sw} is calibrated as an **equivalent pressure**:

$$\epsilon_{sw} \cdot \rho_{sw} \equiv \delta_{sw} \cdot v_{sw}$$

$\rightarrow \rho_{sw} = \delta_{sw} \cdot v_{sw} / \epsilon_{sw} = 0.001 \times 4 \times 10^5 / 0.001 = 4 \times 10^5 \text{ kg/m}^3$ (accretion disk density)

This confirms $\epsilon_{sw} = 0.001$ as the correct coupling for dense stellar wind environments.

8. CP Integration

CP2 update: Add `epsilon_sw = 0.001` to `UQFIBuoyancyCalculator`. Implement `compute_wind_mod(rho_sw, epsilon_sw=0.001)` and apply to all Ubi terms.

Status: ☐ Complete | **CP Stage:** CP2 **Supersedes:** N/A (clarifies δ_{sw} vs ϵ_{sw}) | **Related:** PAPER_064 (4 operational modes, H_SCm), PAPER_086 (Ubi derivation), PAPER_157 (Solar System Ubi)

UQFF computed: Solar wind UQFF correction = $[SSq] \exp(-r/v) = 5.7e-1 \exp(-5.0e-4(1\text{AU}/400\text{km/s})) = 5.7e-1 \exp(-3.2e-3)$ 5.7e-1; dominant at $r < 1\text{AU}$.